NICHOLAS PARNENZINI

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RESEARCH INTERESTS

Compilers, Computer Architecture, GPU Performance Optimization, Efficient LLM Inference

EDUCATION

PhD in Computer Science, Georgia Institute of Technology Aug 2023 – Present

Advisor: Prof. Hyesoon Kim (HPArch Lab). Research on compilers and GPU performance.

Master's Degree in Robotics and Automation Engineering, University of Pisa Oct 2012 – Nov 2015

Final score: 110/110.
Thesis at University of California, Riverside (Mar–Jul 2015): *An Advanced Security Mechanism for Cyber-Physical Systems*.

Bachelor's Degree in Computer Engineering, University of Pisa Sept 2009 – Oct 2012

Final score: 110/110 cum laude.
Thesis: *Hardware-in-the-loop simulator for autopilot based on CAN bus*.

RESEARCH EXPERIENCE

Graduate Research Assistant, HPArch Lab, Georgia Institute of Technology Aug 2023 – Present

- Research on GPU kernel performance optimization at the intersection of compilers and computer architecture.
- **SPLINDER**: a compiler framework for optimizing GPU parallelism via kernel splitting. Paper under review.

WORK EXPERIENCE

Intern, NVIDIA – Deep Learning Performance Architecture (Santa Clara, US) May – Aug 2026

- Analyzed the performance of the Gated DeltaNet (linear attention) inference kernel of Qwen3.5 on Blackwell GPUs, identifying its critical path and occupancy bottlenecks at low batch sizes.
- Derived software and hardware recommendations for future GPU architectures from these bottlenecks.

Intern, NVIDIA – Deep Learning Performance Architecture (Santa Clara, US) May – Aug 2025

- Performance characterization of LLM inference kernels (DeepSeek-V3, Llama 4).

Intern, NVIDIA – Deep Learning Performance Architecture (Santa Clara, US) May – Aug 2024

- Built a tool to investigate kernel launch-latency bottlenecks when using CUDA Graphs on GH200/GB200 systems.

Senior Software Engineer, Arm Ltd – Fast Models (Cambridge, UK) Apr 2021 – Aug 2023

- Design, implementation and testing of functional models for the next generation of Mali GPUs.
- Implemented GPU instructions in C++/LLVM for the Mali functional models, with edge-case test coverage.

Software Engineer, Arm Ltd – Fast Models (Cambridge, UK) Aug 2018 – Mar 2021

- Technical lead for Iris, the debugging tool shipped with the functional models for inspecting simulator state (registers, memory, execution control): feature design, implementation, and bug fixing.
- Designed and developed the backend API for the Fast Models Performance Database, used internally to monitor the simulation performance of models shipped to customers.
- Improved automated test quality by rewriting, fixing, and extending the test suite.

Application Support Engineer, MathWorks Ltd (Cambridge, UK) May 2017 – Aug 2018

- Collaborated on internal feature development projects (C++, MATLAB).
- Developed proof-of-concept examples (ROS/Gazebo, MATLAB) with Application Engineers to demonstrate product value to prospective customers.

- Supported customers with technical issues in MATLAB and Simulink.

Software Engineer, SCISYS GmbH (Bochum, Germany) Mar 2016 – Apr 2017

- Nine-month placement at Thales Alenia Space (Toulouse, France) as Software Verification Engineer: analysis and detection of software anomalies in GACF, a key component of the Galileo Ground Mission Segment.

Apprentice, Embit srl (Modena, Italy) Feb 2016

Contributed to firmware development for embedded devices.

PUBLICATIONS

- **N. Parnenzini**, C. Ahn, H. Pu, S. Jeong, J. Zhao, H. Kim. "SPLINDER: A Compiler Framework for Optimizing GPU Parallelism via Kernel Splitting." *Under review*, 2026.
- C. Ahn, S. Jeong, L. P. Cooper, R. Han, H. Pu, **N. Parnenzini**, J. Zhao, B. Tine, H. Kim. "FastTrackGPU: Static-Analysis-Guided Analytical Modeling for Softcore GPUs." *IEEE Computer Architecture Letters*, 2026.
- S. Jeong, L. P. Cooper, J. M. Lee, H. Choi, **N. Parnenzini**, C. Ahn, Y. Lee, H. Kim, H. Kim. "SparseWeaver: Converting Sparse Operations as Dense Operations on GPUs for Graph Workloads." *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, 2025.
- C. Ahn, S. Jeong, L. Cooper, **N. Parnenzini**, H. Kim. "Comparative Analysis of Executing GPU Applications on FPGA: HLS vs. Soft GPU Approaches." *CGRA4HPC Workshop, IPDPS*, 2024.
- L. Pollini, **N. Parnenzini**, M. Innocenti. "Distributed Real-Time Hardware and Man-in-the-loop Simulation for the ICARO II Unmanned Systems Autopilot." *WSEAS International Conference on Computers*, 2012.

EARLIER RESEARCH PROJECTS

- **Master's thesis** (UC Riverside): a framework modeling an attacker tampering with control inputs of a cyber-physical system, computing the probability of destabilization as a function of the deployed security mechanisms; illustrated on a UAV.
- **iNEMO**: C/MATLAB attitude estimation (roll, pitch, yaw) for an STMicroelectronics low-cost IMU; implemented the Madgwick filter and compared it to ST's reference algorithm.
- **Variable Stiffness Actuators in Gazebo/ROS**: C++ Gazebo plugin simulating qbmove actuators, including their embedded servo-motors. Acknowledged in R. Mengacci et al., *Frontiers in Robotics and AI*, 2021.
- **Bachelor's thesis**: hardware-in-the-loop simulator for the ICARO II autopilot – quadcopter dynamics in C++ embedded in Simulink, CAN bus communication with the microcontroller.

HONORS AND AWARDS

Excellence Award, University of Pisa May 2013

TECHNICAL SKILLS

Languages	C/C++, CUDA, Python, Arm Assembly, Java
GPU / Compilers	LLVM, CUDA Graphs, Nsight Systems/Compute, Triton
Tools & Libraries	SystemC, OpenGL, Git, ROS, Gazebo, MATLAB, Simulink
Spoken languages	Italian (native), English (fluent)